

Al Imam Mohammad Ibn Saud Islamic University
College of Computer and Information Sciences
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Executive Summary

“The current parking system at IMSIU Female Campus (CCIS) causes daily delays, stress, and campus traffic. Students spend time searching for available parking spaces because there is no real-time guidance system. This problem affects students by causing them to be late and increasing traffic on campus, resulting in an inefficient and frustrating experience for students.

To better understand the problem, we surveyed 116 CCIS students. Results showed that 41.1% come daily, and 82.1% spend 5-20 minutes to find a parking space. Also 79.3% of students experience parking difficulties frequently or occasionally. The primary challenges include traffic congestion at entrances and exits 62.4%, lack of available spaces 53.8%, distance of spots from the gate 47.9%, and reserved or closed parking areas 43.6%. Also 69% of students parked in distant or restricted areas due to the unavailability of closer spaces. Multiple solutions were considered for this problem. Expanding parking capacity is hard and time and money consuming. Assigning staff to manage traffic and guide cars is applied but it's considered a temporary solution. Therefore the most effective solution is to implement a smart parking system.

Marakn is a smart parking system that uses Internet of Things (IoT) technology. Magnetic Sensors inserted in the floor of each parking space detect whether the spot is occupied or available, then send data to central system via a low-power wireless network, to intermediate device called Gateway and this device collects signals from sensors and sends them to Cloud Server where data is being analyzed and updated in real time, then displayed in the Markan application on the mobile so that the students can see the available parking in green and occupied in red on an interactive map.

Using Markan will improve the daily parking experience for students in CCIS by providing real-time guidance of available parking. Markn will help to reduce the time and unnecessary traffic spent searching for an empty parking space, which will improve the overall organization and management of parking areas and will promote the use of smart technologies inside the university.”



Introduction

This business report was prepared for the “Business Communication” course under the supervision of Dr. Halah Al-AlShaikh. It addresses the problem that students face in Imam Mohammad Ibn Saud Islamic University, Female Campus, CCIS College, which is the difficulty of finding a parking space. Every morning there are long lines of cars that fill the parking area as students drive around in circles searching for an available space due to the Absence of a smart system. because of the unorganized parking flow students park far away from the entrance or in unauthorized areas. The number of parking spaces is insufficient especially near the entrance. This problem leads to wasting time searching for available spaces, which results in students being late for classes or exams or even an important meeting. adding unnecessary stress to their daily routine.

To address this problem, we introduce Markan, a smart parking meaning an automated parking solution that uses an in-ground smart parking sensor [2]. The system relies on Internet of Things (IoT) technology, which is a network of physical devices equipped with sensors, software, and connectivity that allow them to collect and share data [4]. The system uses Magnetic Sensors that are based on the magnetic flux density to detect the changes in the geomagnetic field and the surrounding magnetic environment [3] to identify whether a parking space is occupied or available. The sensor data is transmitted through a low power wireless network to a Gateway device, a hardware connection point that collects sensor data and transfers it to the cloud [8]. The information is processed in a Cloud Server, a virtual infrastructure hosted by a cloud service provider that stores, processes, and updates data in real time [9]. The system also supports real-time data processing, meaning data is immediately processed and analyzed after it is generated and collected, often within milliseconds [10]. Finally, the Markan mobile application displays an interactive map showing available spaces in green and occupied spaces in red. This solution will help to find a parking space faster and easier and promote the use of technology inside the university.

The scope of Markan is limited to the female campus College of Computer and Information Sciences (CCIS), and targets students who drive their own cars and park on campus. However there are a few limitations: it depends on the sensor's reliability and accuracy, as well as a stable network. The speed of data transfer from the sensor to the application depends entirely on the internet speed and connectivity. The system requires a high first installation cost of sensors, and needs regular maintenance to ensure reliability.

We located the required information and data in this report by using both primary research and secondary research. For the primary research data were gathered through a student survey distributed among the CCIS female students to understand the problem of parking. For the secondary research we gathered information from similar topics and some articles, and previous studies related to smart parking and IoT technology.

This report begins with an analysis of the current parking situation and its effect. Then, the proposed technical solution “Markan” will be described in detail. Lastly we will present the recommendations based on the proposed solution and the analytical studies.

Objectives

- To reduce the time spent searching for an empty parking space.
- To reduce unnecessary traffic inside the campus caused by the search for available parking spaces.
- To improve the overall organization and management of parking areas.
- To promote the use of smart technologies inside the university.



Gap Analysis

To find the main challenges of the current parking system at Imam Mohammad Ibn Saud Islamic University, Female Campus, College of Computer Information Sciences, we do gap analysis to compare the current situation with the desired situation, identify gaps and their causes, and then explain how the Markan application can effectively bridge these gaps.

Table 1 : Gap Analysis

Aspect	current situation	desired situation	Gap	How Markan closes the gap
Parking System	There is no smart or electronic system to identify available parking spaces	The smart system uses IoT sensors, which detect available spaces and display them in real time	Lack of a system to show which parking spaces are free or occupied	Markan will provide a real-time detection through IoT sensors and display availability in the mobile app
Time Management	Students spend a lot of time driving around searching for parking.	Students can find a parking space quickly, easily, and smoothly	Time is wasted, and users often arrive late to classes, meetings, or work.	Markan will help to reduce the time and speed up the process by providing a mobile app that guides the user directly to an available parking space.



Traffic Congestion	Random car movement causes traffic in the college parking lot	smoother movement and traffic-free	The congestion is due to the search for an unorganized parking space	Markan organizes traffic by showing available spots and guiding drivers to it efficiently
User Experience	Parking is exhausting and frustrating process for everyone.	Smooth, stress-free, and non-exhausting experience	Lack of organization and guidance.	Markan provides a clear map to direct guidance on the available position.
Use of Technology	There is no use of technology in the current parking management.	Bring the adoption of IoT and smart systems inside the university	Outdated manual system.	Markan offers a smart system based on Internet of Things technologies to organize parking

From the gap analysis, it shows that the main problem is the absence of a smart system that guides students and displays available parking in real time. The current manual method results in wasted time, traffic on campus, and increased student stress. So with the implementation of Markan, traffic will be significantly reduced, improved efficiency, and improved the daily experience of students parking will be easier and more organized.



Facts, Statistical Evidence, and Trends

To understand what students face in parking challenges daily, we shared an electronic survey using Google Forms and shared it through WhatsApp and Telegram student groups. The survey was open for one week, and we received 117 responses, which provided reliable insights into everyday parking challenges and student opinions for the smart solution.

The results show that attendance at college is high, as 41% of students attend daily and another 41% attend 3–4 days per week, which increases the demand for parking. For the time to find a parking spot, 43.6% of the students needed 5–10 minutes, 38.5% needed 10–20 minutes, and only 13.7% found parking in less than 5 minutes, while 4.3% needed more than 20 minutes. The results also showed that difficulty in finding a parking space is a common problem, with 47% of students facing this challenge frequently and 32.5% occasionally, while 6.8% do not face any difficulty.

For the most challenges that students face while parking, 62.4% is the traffic congestion at the entrances and exits, 53.8% is the lack of available parking spaces, 47.9% is the distance of parking spaces from the gate, and 43.6% is reserved or closed parking areas. Also, 69.2% of students had parked in restricted or distant areas because of the unavailability of closer spaces.

The open-ended responses described further problems, such as being late for lectures or exams, traffic during peak hours, blocked spots because of incorrect parking, and difficulty finding spots near the gate.

After reviewing the challenges, the survey assessed student opinions for the proposed Markan system.

65% of students believed that a smart application displaying available parking spaces would help reduce search time, while 70.1% found a color-coded map useful, and 70.1% believed automatic real-time updates would enhance accuracy. Also, we asked about the most important features 84.5% of students selected showing empty parking spaces, 78.3% want to identify the nearest available spot, and 49.6% want to receive notifications when parking areas become full.



Analysis of Action

After reviewing the current parking situation and evaluating different possible solutions at Imam Mohammad Ibn Saud Islamic University, Female Campus, College of Computer Information Sciences (CCIS), several actions were considered to address the parking challenges:

1- Expanding parking spaces to accommodate more cars.

This solution requires significant financial investment, additional land, and time for construction, which will make it impractical in the short term.

2- Improve the organization of the existing parking system.

by assigning security staff to manage traffic and guide cars.

This solution is currently applied at the university, but it only provides a temporary solution and does not address the main problem, which is the lack of real-time information about available parking spaces.

Also, this manual solution is not always effective, as staff are not always available due to extreme heat and other factors, which limit the reliability of this solution.

3- Implement a smart parking system.

which uses sensors to detect empty spaces and display them in a mobile application (Markan). This solution directly addresses the causes of the problem because it provides real-time visibility, reduces search time, minimizes traffic, and improves the student experience. Based on this analysis, adopting Markan is the most sustainable and efficient solution for the university to enhance parking management and reduce the daily challenges that students face.



Causes of the Problem

The parking problem at Imam Mohammad Ibn Saud Islamic University – Female Campus, College of Computer and Information Sciences (CCIS) is caused by multiple key factors.

- 1- **Absence of a smart system:** This is the most important issue because without providing real-time information about available parking spaces, students must drive around the parking area to find a free spot, which wastes time and increases traffic congestion inside the campus.
- 2- **Unorganized parking flow:** There is no clear structure to manage traffic or guide cars. Students move in different directions, some cars take more than one space, and sometimes security staff try to manage manually, but it's not reliable since they are not always available; this makes the parking process unorganized
- 3- **Insufficient number of parking space near the main entrances:** Many of the parking are located far from the college gates, making it difficult for students to access the buildings easily, especially during hot weather or when they are in a hurry.



Proposed IT Solution

Markan system aims to improve the experience of parking within the College of Computer and Information Sciences, Female Campus of Imam Muhammad bin Saud Islamic University, by employing Internet of Things (IoT) technologies to link intelligent system elements together in an integrated network. The system relies on the use of magnetic sensors that are installed in the floor of each parking lot to measure the change in the magnetic field when there is an object such as a car, allowing the system to determine whether the parking is busy or empty with a high level of accuracy. This type of sensor was chosen because of its high efficiency in outdoor environments compared to sound sensors, and because it is actually used in smart parking projects in Saudi Arabia. Reports also show that the magnetic sensor market in the Kingdom is growing due to the rise of smart city applications and smart parking systems, which makes it a practical and realistic option. [11]

These magnetic sensors then send the data to the central system through a low-power wireless network. This type of network is known for its low energy consumption and its ability to cover large areas efficiently. [12]

Such wireless technologies are widely used in Saudi smart city projects. [7]

After that, the data is collected from the sensors by an intermediate device called a Gateway. The Gateway gathers the signals from all nearby sensors and prepares them before sending them to the cloud. [8]

Next, the Cloud Server receives the data from the Gateway, analyzes it, and updates the status of each parking spot in real time. [9]

After processing, the updated information appears in the Markan mobile application. The student can easily see the available parking (green) and the busy ones (red) on an interactive map. The system works in an integrated and organized sequence—from the magnetic sensor, to the Gateway, then to the cloud server, and finally to the application. This allows the data to be updated instantly without any manual intervention and helps achieve the main goal of the project, which is to facilitate the parking process, reduce congestion, and improve the daily experience of female students inside the college.



Procedures / System Process

Markan system works in a professional way that starts from the moment the car enters the parking area and ends with displaying the parking status in the application on the student's mobile. The process goes through several interrelated stages as follows:

1- Entering the car into the parking:

When the car approaches a parking spot in the college, the magnetic sensor installed in the parking floor captures the change in the magnetic field caused by the presence of a large metal object such as the car, so the system immediately recognizes that the parking has become busy.

2- Transmitting data over the wireless network:

After detecting the change, the sensor sends a signal to the Low Power Wireless Network, which transmits data efficiently without significant energy consumption, the same technology used in modern Internet of Things (IoT) systems. [12]

3- Signal transfer to the Gateway:

The Gateway receives these signals from multiple sensors in the same area and initially assembles and processes them before sending them to the cloud server.

The Gateway acts as an intermediary between local sensors and the Internet because it connects small devices like sensors. [6]

4- Data processing in the cloud server:

The cloud server receives data from the Gateway, then analyzes it and updates the status of each parking spot in the database. [9] If the parking changes from empty to busy or vice versa, it is updated in real time.

5- View status in the Markan application:

Finally, the updated data arrives on the Markan application on students' phones.

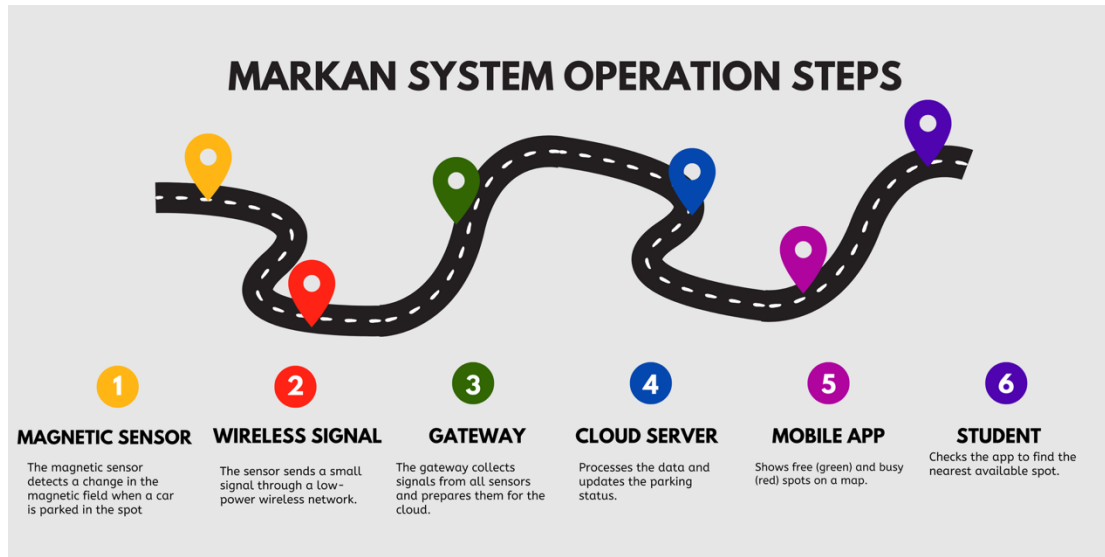


Figure 1 : Markan System Operation Steps (Summary)

Methods and Approaches

Markan system relies on Internet of Things (IoT) technologies that connect smart devices to the cloud server for real-time data exchange. Magnetic sensors were chosen because they are accurate in car detection and are not affected by external conditions such as heat or dust, unlike acoustic sensors. A low-power wireless network such as NB-IoT has also been used to reduce battery consumption and ensure wide coverage. [12] This type of technology is actually used in Saudi projects such as NEOM, making the system practical and applicable in the university environment. [7]

Pros and Cons

Markan system has many good features that make it helpful and easy to use, but like any smart system, it can also face some small challenges during work. The following points explain both the positive and negative sides of the system in a simple way:

Pros:

The system is characterized by its high accuracy in detecting busy and empty parking using magnetic sensors that work well even in different outdoor conditions for example the dust and rain.

- It updates the information directly in the application, so students can know where the available parking is at the same moment without wasting time searching randomly.
- This helps reduce congestion and save time and effort for female students when they enter or leave the college.
- The system is also simple and easy to understand, making it comfortable for students to use.

Cons:

- The system may face some challenges such as the high first installation cost of sensors and the need for regular maintenance to keep them working well.
- also there is possibility of weak network coverage in some areas or temporary technical problems, because the wireless signal inside the parking area may not always be strong everywhere.

But with good follow-up and good planning, these problems can be fixed and Markan can stay an effective and helpful system for students.



Markan



Criteria

The Markan system was chosen based on criteria that make the solution practical and suitable for the College of Computer and Information Sciences (CCIS) environment. One of the most important of these criteria is ease of application the solution without the need for major changes in the structure of current parking also the realism of solution where the same technology is used in existing Saudi Arabia projects, in addition to being scalable in the future for example coverage of the entire university or adding notifications for users. The solution is also effective in reducing congestion and improving the experience of female students just like smart parking systems used in other universities[1], which is the main goal of the markan system .



Findings and Discussion

After the survey analysis it indicates the strong need for parking among students who regularly attend the university. First, about four out of five students reported attending at least three days per week, which puts a heavy pressure on the limited parking spaces available. And because of this, finding a free spot often takes between 5 to 20 minutes. Also the data shows that most traffic grows at the entrances and exits, since many cars are waiting to enter or leave at the same time. This current situation is considered as one of the biggest challenges. This resulted in forcing many students to park far away or in restricted areas, which led to delays, frustration, and extra stress before classes.

These findings clarified that the current parking management system does not meet student needs. The long search times and crowded areas show problems not only with the number of spaces but also with how traffic is managed.

Previous studies, such as at Universiti Utara Malaysia, reported that weak parking systems caused delays and stress for students [13]. In the same way, many students here end up parking in distant or private areas, which increases walking time and exposes them to heat, especially during summer. This shows the absence of a real-time system that can guide drivers to available spaces.

A different study at King Fahd University of Petroleum & Minerals (KFUPM) about the implementation and testing of a smart parking system as a case study. It utilizes sensors and digital technology to direct drivers to free parking spots. This system gives evidence advantages, such as lower wasted time, and stress levels, also enhancing the student satisfaction [14]. This example shows that smart parking solutions are not just theoretical but can be implemented in real university environments, and serves as a viable model for universities facing similar issues.

In order to solve these challenges, we introduced a Markan smart parking system aimed at delivering real-time updates, a color-coded map showing the available spaces, and alerts when certain areas are full. These features directly eliminate the issues students pointed out, like wasting too much time searching and not knowing where the closest available spot is. Providing clear and real information will help in reducing congestion, improving punctuality, and make smoother traffic movement throughout the campus.



Benefits

- Speed up the process of finding parking spaces, which reduces the total time students spend searching.
- Enable students to be more punctual, which lowers stress and gives them extra time for important tasks.
- Identify peak times and locations of traffic congestion in parking areas.
- Reduce overall traffic congestion, leading to lower emissions and a cleaner environment.
- Improve students' daily experience and satisfaction.
- Strengthen CCIS College's institutional reputation as a leader in innovation.
- Support Saudi Vision 2030 goals by contributing to the development of a smart campus.

Recommendations

1. Implement a pilot program in the most congested areas to test the system, validate sensor accuracy, ensure network stability, and observe how users interact with it.
2. Expand the system to cover more parking areas, other colleges, and over time the entire university.
3. Integrate the system with other transportation services, such as bus arrival time tracking.
4. Add an alert feature to notify users when congestion occurs or when parking areas are full.
5. Perform regular checks and updates on system performance, including sensors and network components.
6. Collect student feedback to identify gaps and make necessary improvements.
7. Publish clear announcements to inform students about the new system, highlight its benefits, and encourage usage.



Action Items

To make the implementation of the Markan system structured and easy to follow, the following table has been defined with the main action items:

Table 2 : Action Items

ID	Action Description	Assigned To	Priority	Due Date	Status
AI-1	Conduct sensor testing in the most congested parking area	IT Team	High	30-Jan-26	In Progress
AI-2	Install the smart parking system across all CCIS parking areas	Facilities Team	High	20-Apr-26	Not Started
AI-3	Connect the system to the cloud for real-time updates	App Team	Medium	25-Jun-26	Not Started
AI-4	Organize training sessions to guide students on system usage	HR+IT	Medium	15-Aug-26	Planned
AI-5	Monitor system performance and maintain data accuracy	Project Group	High	Ongoing	Continuous



Conclusions

The current parking system is temporary, manual, and inefficient. It causes congestion, long search times, and frustration for students. Many are forced to park far away or in restricted areas, which adds stress and delays. Expanding physical parking capacity is not practical, and without change, these problems will continue to grow as the number of students increases.

The solution is to adopt a smart IoT-based system. With magnetic sensors, gateways, and cloud servers, the Markan system can provide real-time updates, guide drivers to available spots, and reduce congestion. This modern approach replaces uncertainty with accurate information, helping students save time and making traffic flow smoother across campus.

In summary, the Markan system is more than just a fix for parking. It represents a technological transformation for the university, aligned with Saudi Vision 2030, and shows how smart solutions can improve everyday student life. The system will make students more punctual, reduce stress, and improve satisfaction. And strengthen the university's reputation as a leader in innovation and provide a scalable model that can be expanded to other colleges and campuses in the future. Finally, it turns a daily challenge into an opportunity for efficiency, sustainability, and progress.



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Markan



Appendixes

Appendix A: Survey Results

The survey was divided into two parts: Part 1 focused on the current situation and challenges, and Part 2 focused on proposed solutions.

Part 1: Current Situation and Challenges:

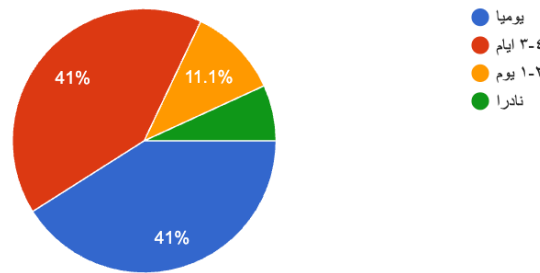
Question 1: How many times per week do you attend college?

(كم مرة في الأسبوع تحضرين إلى الكلية؟)

نسخ الرسم البياني

كم مرة في الأسبوع تحضرين إلى الكلية؟

117 ردًا



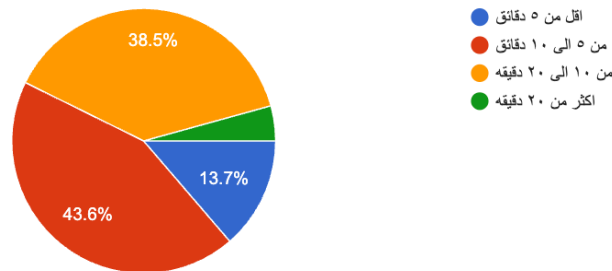
Question 2: How much time does it usually take you to find a suitable parking spot on campus?

(كم من الوقت يستغرقك عادةً العثور على موقف مناسب داخل الكلية؟)

نسخ الرسم البياني

كم من الوقت يستغرقك عادةً العثور على موقف مناسب داخل الكلية؟

117 ردًا



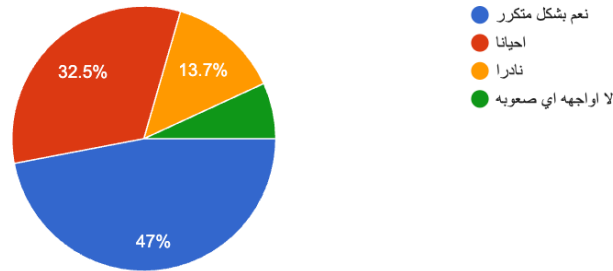
Question 3: Do you face congestion or difficulty finding a parking spot when you arrive at the college?

(هل تواجهين ازدحامًا أو صعوبة في إيجاد موقف عند قدومك للكلية؟)

نسخ الرسم البياني

هل تواجهين ازدحامًا أو صعوبة في إيجاد موقف عند قدومك للكلية؟

117 ردًا



Question 4: What are the main difficulties you face while trying to park ?

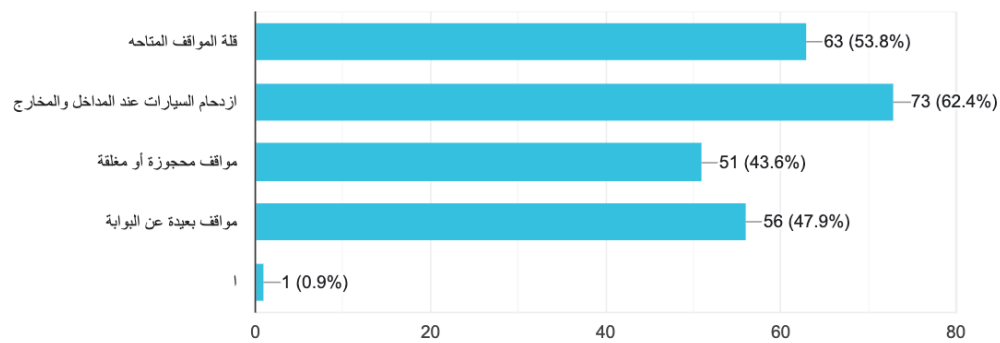
(ما أبرز الصعوبات التي تواجهينها أثناء محاولة الركن؟)

نسخ الرسم البياني

ما أبرز الصعوبات التي تواجهينها أثناء محاولة الركن؟

(يمكن اختيار أكثر من إجابة)

117 ردًا



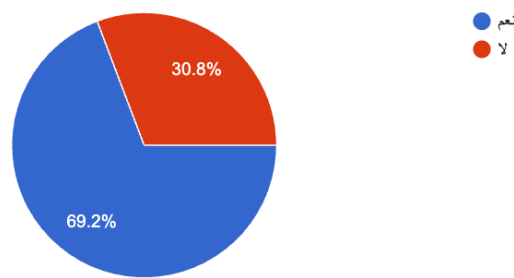
Question 5: Have you ever had to park in a forbidden or very far place due to the lack of available spots?

(هل اضطررت يوماً للوقوف في مكان ممنوع أو بعيد جداً بسبب عدم توفر المواقف؟)

نسخ الرسم البياني

هل اضطررت يوماً للوقوف في مكان ممنوع أو بعيد جداً بسبب عدم توفر المواقف؟

117 ردًا



Question 6: Please describe a situation where you faced difficulty finding a parking spot on campus

(يرجى وصف تجربة واجهت فيها صعوبة في العثور على موقف داخل الكلية)

يرجى وصف تجربة واجهت فيها صعوبة في العثور على موقف داخل الكلية:

13 ردًا

موجود مواقف بس صعب نحصلها لانها تصير بين سيارتين

محد يعرف بيركن

ازدحام غير طبيعي

الطلاب لا يعرفون كيف يقفون سياراتهم

.

كان عندي اختبار ومستعجله ومالقيت موقف قريب من البوابه الا حق ذوي الاحتياجات الخاصة واضطريت اوقف فيه واخذت مخالفه

اذا جيت بدري للجامعة القى مواقف

الزحمة تكون عند اوقات بدء وانتهاء المحاضرات

تاخرت عن اختبار عشان ادور على موقف

Part 2: Proposed Solutions

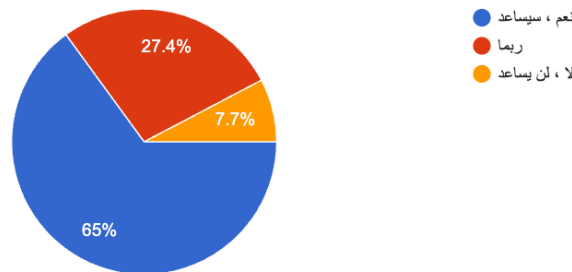
Question 7: To what extent do you think a smart application showing available spots would help reduce the time spent searching for a parking spot?

(الى أي مدى ترين أن وجود تطبيق ذكي يوضح المواقع المتاحة سيساعد في تقليل الوقت المستغرق للبحث عن موقف؟)

نسخ الرسم البياني

الى أي مدى ترين أن وجود تطبيق ذكي يوضح المواقع المتاحة سيساعد في تقليل الوقت المستغرق للبحث عن موقف ؟

117 ردًا



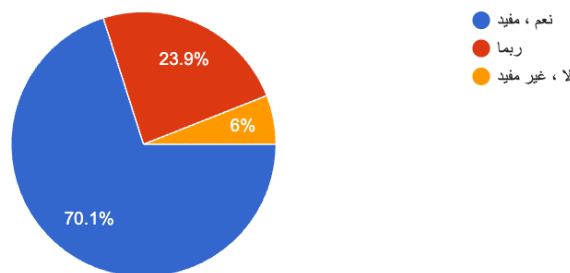
Question 8: Do you think having an interactive map in the application showing occupied and empty spots with colors would be useful to you?

(هل ترين أن وجود خريطة تفاعلية في التطبيق تُظهر المواقع الفارغة والمشغولة بالألوان سيكون مفيدًا لك؟)

نسخ الرسم البياني

هل ترين أن وجود خريطة تفاعلية في التطبيق تُظهر المواقع الفارغة والمشغولة بالألوان سيكون مفيدًا لك؟

117 ردًا



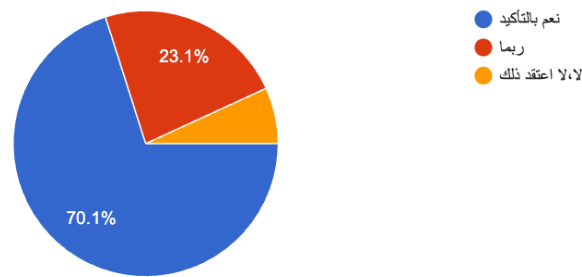
Question 9: Do you think automatic updates of parking availability in the app would make the system more accurate and easier to use?

(هل تعتقد ان التحديث التلقائي لحالة المواقف في التطبيق سيجعل النظام أدق وأسهل في الاستخدام؟)

نسخ الرسم البياني

هل تعتقد ان التحديث التلقائي لحالة المواقف في التطبيق سيجعل النظام أدق وأسهل في الاستخدام؟

117 ردًا



Question 10: Which feature do you consider the most important in the application?

(ما الميزة التي تعتبرينها الأكثر أهمية في التطبيق بالنسبة لك؟)

نسخ الرسم البياني

ما الميزة التي تعتبرينها الأكثر أهمية في التطبيق بالنسبة لك؟

115 ردًا

